Theory-based priors for the output gap

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Abstract

Estimating potential output and the output gap requires identifying restrictions on the dynamics of trend and cycle. We develop a general econometric framework that specifies theory-based restrictions directly over the joint distribution of the latent trajectory, rather than through a particular state evolution equation. The approach nests standard unobserved-components models, while allowing external quantitative or qualitative information to be incorporated in a transparent and precision-weighted manner. Simulation evidence suggests that our prior can recover potentially asymmetric cyclical evolutions without having to estimate potentially challenging non-linear models. An empirical application to U.S. GDP shows that the Covid pandemic was not associated with a drop in the potential level of output, but the Great Financial Crisis was.

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1 Introduction

Decomposing GDP into potential output and the output gap is central to macroeconomic analysis and policy design. Estimates of the output gap guide stabilization policy and forecasting, while potential output anchors assessments of long-run growth and fiscal sustainability. Because neither component is directly observable, their measurement hinges on identifying assumptions. The choice and implementation of these assumptions are not innocuous: they shape inference and ultimately policy conclusions.

At a broad level, the literature offers two widely used approaches. The production-function approach defines potential output as the level of output consistent with current technologies and normal utilization of capital and labor, typically constructing it from estimates of trend inputs and total factor productivity (Fleischman and Roberts, 2011). The statistical approach—most prominently unobserved-components (UC) models—defines potential output as the stochastic trend of GDP (Jarocinski and Lenza, 2018). There are other structural approaches, such as DSGE models, but in practice most policy institutions rely primarily on either production-function or statistical decompositions. These approaches differ mainly in how identifying restrictions on trend and cycle are imposed. See Canova (2025) for a comprehensive analysis of alternative existing methods.

This paper improves upon the existing approaches by developing a general econometric framework for incorporating external restrictions on latent trend–cycle decompositions in a transparent and continuously tunable manner. The key innovation is to specify identifying restrictions directly at the level of the joint distribution of the latent trajectory, rather than through a particular state evolution equation. Specifically, we formulate priors over the entire sample path of potential output and the output gap, parameterized by a mean vector and covariance matrix.

This formulation nests standard unobserved-components models as a special case: conventional specifications such as random-walk trends or local linear trends correspond to particular covariance structures implied by state-space transition equations. Our approach therefore generalizes the UC framework by decoupling identification from a specific parametric law of motion and instead allowing restrictions to be imposed directly on admissible trajectories.

Econometrically, the framework exploits the joint Gaussian representation of the latent states, yielding closed-form expressions for posterior inference and a precision-

weighted combination of information from the statistical likelihood and external sources. Implementation requires only the specification of a mean and covariance structure for potential output and the output gap. These can be obtained from simulations of structural models, auxiliary statistical models, survey-based assessments, or other sources of prior information.

The shift from process-level restrictions to distributional restrictions over the full path enlarges the set of admissible dynamics relative to standard UC models while preserving tractability and interpretability. Because restrictions are imposed directly on the joint distribution of the latent trajectory, the framework permits the incorporation of structural implications, qualitative beliefs about business-cycle features (such as duration, asymmetry, or volatility), or judgmental assessments in a transparent and precision-weighted manner. For example, if a structural model implies that expansionary business-cycle fluctuations are shallow and last eight quarters while recessions last four quarters and are abrupt and deep, these beliefs can be encoded directly in the covariance structure of the output gap without estimating the structural model itself. One can simply simulate from the complex non-linear model to inform prior moments in the more tractable Gaussian model, rather than estimate it. See also Loria et al. (2022) for a similar approach within a different class of models. Quantitatively, this flexibility allows the resulting estimates of potential output and the output gap to depart from those implied by conventional UC specifications whenever external information provides sufficiently precise restrictions, thereby making explicit how alternative identifying assumptions affect inference.

Although we focus on potential output and the output gap, the proposed framework applies more generally to the estimation of latent "star" variables, providing a unified approach to incorporating external information into trend–cycle decompositions.

To assess the properties of the proposed framework, we first conduct a set of simulation experiments. First, we show that, indeed in this class of model, prior beliefs matter considerable when estimating potential output and gaps from the data. This suggests that there is value in retaining as much flexibility in the prior beliefs as possible, as made possible within our framework. Then we show that our Gaussian prior specified directly on the latent variable of the output gap can uncover asymmetric business cycle fluctuations without having to estimate a non-linear model featuring asymmetrics, which would be computationally more demanding. Rather, it only requires simulating from a model featuring non-linearities in order to compute the mean and variance of the prior on the cycle. By comparison, the standard AR(2) prior with

oscillatory parameters performs poorly in recovering the true unobserved business cycle fluctuations.

We then apply the framework to U.S. real GDP. We use our methodology to study if the drop in US real GDP during the Great Financial Crises and during the Covid pandemic signal a decline in potential output, or only a widening of the output gap. We use prior beliefs that are informed by widely-held views about the business cycle, for instance that the prior distribution on the trend should only allow for a relatively tight variance around the mean, contrary to explosive non-stationary patterns from the random walk with drift model. Similarly, we formulate prior beliefs on the business cycle that are centered at zero, but feature a covariance structure that introduces the beliefs that economic expansions last longer, while recessions can be sharper and deeper. Last, we widen the prior variance of both the trend and the cycle in the years of the financial crisis and the pandemic, in order to potentially capture such events via the model rather than via the measurement error. We find evidence that the Great Financial Crisis was associated with an average annualized growth of potential output that dropped from an average 2.1% to as little as 0.2%, recovering the pre-crisis growth rates only in 2012. By contrast, we find that the Covid pandemic only led to a temporary drop in output, with an unchanged level of potential output.

The paper relates to a vast literature. Several papers have documented the standard properties of business cycle dynamics, dating back to Burns and Mitchell (1946), and including more recent work, for instance Stock and Watson (2002) and Stock and Watson (2013). Our method provides a way to introduce the qualitative business cycle features from this literature into a tractable joint distribution. As, for instance, Grant and Chan (2017), Planas et al. (2008), and Hasenzagl et al. (2022) we take a Bayesian approach to output gap estimation, although a large frequentist literature explores very related concepts (Harvey, 1985). We use precision sampling for inference (Chan and Jeliazkov, 2009). More broadly, our method can be applied to any 'star' decomposition, including for instance to r star (Del Negro et al., 2017, 2019). Last, several work has been done to shape prior distributions for time series methods using theory-based information, see for instance Ingram and Whiteman (1994), Del Negro and Schorfheide (2004) and Renzetti (2025). The contribution of our paper is similar in spirit, and works in the field of latent variable estimation.

The remainder of the paper is organized as follows. Section 2 presents the econometric framework and establishes its relationship to standard state-space representations. Section 3 reports simulation evidence. Section 4 applies the methodology to U.S.

GDP and compares the resulting estimates with conventional approaches. Section 5 concludes.

2 The model

For ease of exposition, we begin by considering a simple model for log GDP, y_t:

$$y_t = \tau_t + g_t + u_t, \qquad u_t \sim \mathcal{N}(0, \sigma_u^2), \tag{1}$$

where τ_t is the trend component (potential output), g_t is the cyclical component (the output gap), and u_t is a measurement error term, assumed to be Normally distributed with mean zero and variance σ_u^2.

Our objective is to estimate τ_t and g_t, that is, potential output and the output gap, which must be inferred from the observed data on y_t. Importantly, without additional assumptions, these two objects cannot be separately identified. Any decomposition of output into trend and cycle necessarily requires further structure, since multiple combinations of τ_t and g_t can generate the same time series y_t and are observationally equivalent from the perspective of the likelihood.

Within a Bayesian framework, inference on τ_t and g_t in model (1) relies on specifying prior distributions through assumptions on their respective laws of motion. These laws of motion encode our beliefs about the dynamic properties of each component. Typically, the trend component is modeled as a slowly evolving process—reflecting the idea that potential output changes gradually over time—while the cyclical component is modeled as a stationary and mean-reverting process, capturing short- to medium-run fluctuations around the trend. Identification is therefore achieved by imposing distinct dynamic structures - expressed as different priors - on the two latent processes.

In this setting, it is not possible to adopt a fully uninformative prior, as the likelihood alone cannot disentangle the two latent components, hence posterior inference is necessarily shaped by the prior assumptions on their dynamics. In other words, because the model is not identified in the absence of restrictions, the prior necessarily plays a central role in the estimation. The way in which movements in observed output are allocated between the trend and cyclical components depends directly on these prior assumptions.

These considerations motivate our approach: we aim to retain the traditional purely statistical specification - where the trend and cyclical components are mod-

eled as standard stochastic processes, such as a random walk for the trend and an autoregressive process for the cycle - as a benchmark or backstop. At the same time, we allow for the incorporation of additional beliefs when formulating the priors. In this way, the framework remains anchored in conventional unobserved-components models while providing flexibility to depart from them when warranted.

For expositional clarity, we focus on a univariate setting. However, much of the literature estimates τ_t and g_t within multivariate models, where they are interpreted as common factors driving both nominal and real variables. See for example Jarocinski and Lenza (2018); Hasenzagl et al. (2022). In such frameworks, output is embedded in a larger vector y_t, and identification exploits cross-variable co-movements. The arguments developed here can be generalized to that broader setting.

The role of the measurement error term u_t also merits attention. In the baseline specification, it is assumed to be Normally distributed, which facilitates estimation within a standard state-space framework. Nevertheless, the conceptual discussion can be extended to accommodate non-Gaussian measurement errors without altering the main insights.

Finally, while this paper focuses primarily on introducing theory-based beliefs on the output gap g_t, the same ideas can be applied to the trend component τ_t. The framework developed in the next section extends naturally to the specification of priors governing the dynamics of the trend process as well.

2.1 Traditional approach using statistical models

In the traditional statistical approach, Bayesian estimation of model (1) relies on specifying prior laws of motion for the latent components τ_t and g_t. A common specification in the literature starts from the following prior laws of motion:

$$\tau_t = \mu + \tau_{t-1} + \eta_t^\tau, \qquad \eta_t^\tau \sim \; iid \; N(0, \sigma_\tau^2), \tag{2}$$

$$g_t = \phi_1 g_{t-1} + \phi_2 g_{t-2} + \eta_t^g, \qquad \eta_t^g \sim \; iid \; N(0, \sigma_g^2). \tag{3}$$

Under this specification, the output trend evolves as a random walk with drift, capturing its slow-moving nature, while the output gap follows an AR(2) process. The autoregressive coefficients (ϕ_1, ϕ_2) are typically restricted to lie in the stationary and oscillatory region of the parameter space, reflecting the idea that the output gap displays cyclical, mean-reverting dynamics. It is worth emphasizing that this specification is only one possible benchmark. The literature has considered a variety of alternative

purely statistical priors, based on different stochastic processes for both the trend and the cyclical component, including local level or local linear trend models for the trend and alternative autoregressive or higher-order dynamic specifications for the cycle (see, for instance, Grant and Chan, 2017, Planas et al., 2008).

Bayesian estimation proceeds by introducing priors on the underlying parameters. Let $\boldsymbol{\theta} = (\boldsymbol{\theta}_g', \boldsymbol{\theta}_\tau')'$ denote the full parameter vector, where $\boldsymbol{\theta}_g = (\phi_1, \phi_2, \sigma_g^2, g_0, g_{-1})'$, $\boldsymbol{\theta}_\tau = (\mu, \sigma_\tau^2, \tau_0)'$. The prior structure can be summarized as

$$\begin{aligned} p(\boldsymbol{\tau}, \boldsymbol{g}, \boldsymbol{\theta}) &= p(\boldsymbol{\tau}, \boldsymbol{g}|\boldsymbol{\theta})\ p(\boldsymbol{\theta}), \\ &= p(\boldsymbol{\tau}|\boldsymbol{\theta}_\tau)\ p(\boldsymbol{\theta}_{\boldsymbol{\tau}})\ p(\boldsymbol{g}|\boldsymbol{\theta}_g)\ p(\boldsymbol{\theta}_{\boldsymbol{g}}). \end{aligned} \tag{4}$$

This prior is hierarchical in the sense that it factorizes into separate components for the output trend and the output gap, conditional on their respective parameter vectors and initial conditions. Each process has its own set of parameters governing its dynamics and initial conditions. Conditional on $\boldsymbol{\theta}_\tau$ and $\boldsymbol{\theta}_g$, the latent processes $\boldsymbol{\tau}$ and $\boldsymbol{g}$ are independent, and $(\boldsymbol{\theta}_\tau, \boldsymbol{\theta}_g)$ are also independent. Both assumptions can be relaxed.

Focusing on the output gap, the parameters $(\phi_1, \phi_2, g_0, g_{-1})$ determine the conditional mean dynamics of the process, while $(\phi_1, \phi_2, \sigma_g^2)$ determine its autocorrelation structure and overall variability. Conditional on these parameters, the $T \times 1$ vector $\boldsymbol{g}$ follows a multivariate Normal distribution:

$$\boldsymbol{g}|\phi_1, \phi_2, g_0, g_{-1}, \sigma_g^2 \sim \mathcal{N}\left(\boldsymbol{\mu}_g(\phi_1, \phi_2, g_0, g_{-1}), \sigma_g^2 \boldsymbol{V}_g(\phi_1, \phi_2)\right), \tag{5}$$

where the mean and covariance matrix are functions of the autoregressive coefficients, innovation variance, and initial conditions. In particular, following for instance Grant and Chan (2017), the mean can be written as

$$\boldsymbol{\mu}_g = \boldsymbol{H}_\phi^{-1} \boldsymbol{c}_\phi, \tag{6}$$

and the covariance matrix as

$$\boldsymbol{V}_g = \boldsymbol{H}_\phi^{-1} \boldsymbol{H}_\phi^{-1'}, \tag{7}$$

where $\boldsymbol{H}_\phi$ is the $T \times T$ lower-triangular matrix encoding the AR(2) structure, with

$$\boldsymbol{H}_\phi = \begin{bmatrix} 1 & 0 & \dots & 0 \\ -\phi_1 & 1 & \dots & 0 \\ -\phi_2 & -\phi_1 & 1 & 0 \\ 0 & \ddots & -\phi_1 & 1 \\ 0 & 0 & -\phi_2 & -\phi_1 \end{bmatrix} \quad \boldsymbol{c}_\phi = \begin{bmatrix} \phi_1 g_0 + \phi_2 g_{-1} \\ \phi_2 g_0 \\ 0 \\ 0 \\ 0 \end{bmatrix}. \tag{8}$$

Within this basic framework, posterior inference is typically conducted using a Gibbs sampling algorithm to simulate from the joint posterior distribution. Conditional on the parameters, the latent trajectories τ_t and g_t can be drawn using standard filtering and smoothing methods for linear Gaussian systems (Carter and Kohn, 1994). Alternatively, one may exploit a precision-based sampling approach and draw the entire vector of latent variables jointly, as in Grant and Chan (2017). Both strategies are computationally convenient in this linear Gaussian setting and allow efficient exploration of the posterior distribution.

Finally, it is worth emphasizing that this approach is purely statistical: identification of the output gap relies on the assumed stochastic specification and the associated prior distributions over its parameters, given the prior specification for the trend component. If one views the chosen statistical process as an adequate representation of cyclical dynamics, the framework is internally coherent. However, the resulting estimation of the output gap is tightly constrained by that assumption. Its persistence, amplitude, and overall dynamic properties are determined by the imposed process and its parameter priors. Within this setup, it is not possible to incorporate alternative beliefs about the timing, shape, or qualitative features of the output gap without modifying the prior law of motion itself.

2.2 Our prior for the output gap

In this section, we introduce beliefs about the output gap that go beyond those that can be imposed solely through a statistical specification of its prior law of motion, such as in (2). The key idea is to enrich the prior structure in a way that allows for additional, potentially theory-driven information to enter the estimation.

What kinds of beliefs might we want to incorporate? Business cycle models, for example, often imply specific properties of cyclical fluctuations that we may wish to take into account. These may include restrictions on the typical duration of business

cycles, their degree of persistence, or the presence of oscillatory dynamics. Such beliefs can be captured by oscillatory AR(2) processes with restrictions on their oscillation, see Planas et al. (2008). Yet, more generally, theory may also suggest asymmetries between expansions and recessions, state-dependent dynamics, or other qualitative features of cyclical behavior that are difficult to encode within a simple autoregressive structure. Such considerations motivate a more flexible prior specification.

Our main insight is that specifying a prior directly for the entire vector $\boldsymbol{g}$ can provide additional flexibility relative to restricting attention to a parametric law of motion such as the AR(2) process in (2). By moving from process-level priors to a distribution over the full trajectory of the output gap, we gain the ability to incorporate richer beliefs about its joint behavior over time.

To illustrate the idea, suppose we have a structural or theoretical model that generates a complex process for the output gap. The model may feature nonlinearities, state dependence, or other mechanisms that produce dynamics not easily captured by a low-order autoregressive representation. In principle, one could replace equation (2) with this more complex model and accept that inference will most likely be more challenging. We argue that instead of estimating this more complex model on the data, one can use it in simulation to extract useful information to form the prior mean and covariance structure (or other moments) of cyclical fluctuations. Equation (5) suggests that we can inform the estimation of the output gap by exploiting a theory-implied mean function and covariance matrix. In practice, we can simulate the theoretical model, construct the empirical mean vector and covariance matrix of the implied output gap process, and use these objects -denoted by $\boldsymbol{\mu}_{TC}$ and $\boldsymbol{\Sigma}_{TC}$ - to center our prior for $\boldsymbol{g}$. In this way, the prior reflects the theoretical properties of the model without requiring full structural estimation.

In practice, we would like theory-based beliefs to influence the estimation of the output gap without fully determining the prior or dominating the inference on g_t. The goal is to allow theory to inform the prior in a disciplined way, while retaining the flexibility of a statistical benchmark. Motivated by this consideration, we specify the prior for $\boldsymbol{g}$ as

$$p(\boldsymbol{g} \mid \boldsymbol{\theta}) \sim \mathcal{N}\left(\boldsymbol{\mu}^{\star}, \sigma^2 \boldsymbol{\Sigma}^{\star}\right). \tag{9a}$$

$$\boldsymbol{\mu}^{\star} = \boldsymbol{\Sigma}^{\star}\left(\boldsymbol{\Sigma}_{STAT}^{-1}\, \boldsymbol{\mu}_{STAT} + \lambda\, \boldsymbol{\Sigma}_{TC}^{-1}\, \boldsymbol{\mu}_{TC}\right), \tag{9b}$$

$$\boldsymbol{\Sigma}^{\star} = \left(\boldsymbol{\Sigma}_{STAT}^{-1} + \lambda \boldsymbol{\Sigma}_{TC}^{-1}\right)^{-1}. \tag{9c}$$

This specification combines the purely statistical prior with the theory-consistent prior

in a precision-weighted fashion. The subscript $STAT$ refers to the portion of the prior derived from the purely statistical specification of the output gap. In the benchmark example considered above, where the gap follows an AR(2) process, we have $\boldsymbol{\mu}_{STAT} = \mu_g$ and $\boldsymbol{\Sigma}_{STAT} = V_g$.

The parameter λ governs the relative weight assigned to the theory-based component. It provides a transparent and continuous mechanism for controlling how strongly theory-based beliefs affect the estimation of the output gap. When $\lambda = 0$, the prior collapses to the standard statistical specification commonly used in the literature. As λ increases, the mean and covariance structure implied by the theoretical model exert greater influence on the prior for $\boldsymbol{g}$. For $\lambda = 1$, $\boldsymbol{\Sigma}_{STAT}^{-1} = 0$, $\sigma^2 = 1$, the model collapses to

$$p(\boldsymbol{g} \mid \boldsymbol{\theta}) \sim \mathcal{N}\left(\boldsymbol{\mu}_{TC}, \boldsymbol{\Sigma}_{TC}\right). \tag{10}$$

which only relies on theory-based beliefs.

Conceptually, the prior distribution is constructed as the normalized product of two independent Gaussian sources of information about $\boldsymbol{g}$, sharing a common scale parameter σ_g^2 : one capturing the AR(2) dynamic structure in (5), and the other reflecting theory-based qualitative external information. It is this feature that safegards the Normality of the composite prior distribution. Inference with (10) is straightforward, as the conditional posterior of $\boldsymbol{g}$ is jointly Normal. Inference with (9) requires sampling numerically from the conditional posterior of the underlying hyperparameters.

3 Simulation

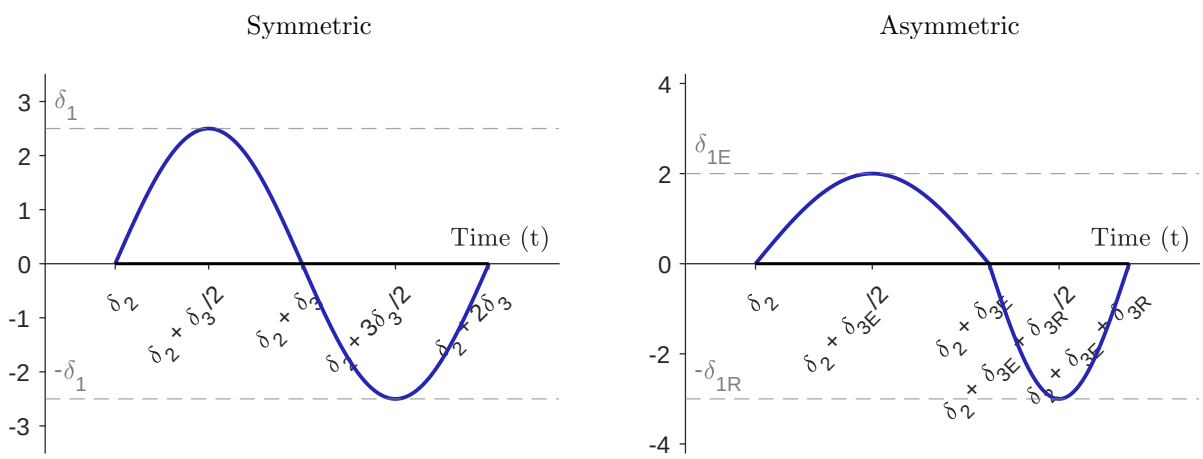
A simulation on pseudo data helps illustrate that prior beliefs matter considerably in the estimation of latent variables, due to lack of identification in this class of models. This suggests that there is merit in retaining as much flexibility as possible in the formulation of prior beliefs, as argued in subsection 2.2.

We consider two datasets, both of size $T = 100$. For both datasets we simulate the true trend as the realization of a random walk with drift, using the same realization in order to ensure the same trend for both datasets. Starting from equation (2), we set $\tau_0 = 0$ and set μ such that the expected value of the trend equals 100 for the last period. We set $\sigma_\tau^2 = 0.05$ as the variance of the error term in the trend process.

The first dataset simulates the cycle as the realization of an AR(2) process. Starting from equation (3), we set the parameters to imply that the process is stationary,

oscillatory, and with periodicity of 8 periods. This means that the response of the process to an impulse of positive sign takes 8 periods to complete a full expansion and then 8 periods to complete a full recession. The modulus of the oscillatory process is set to 0.90, which ensures that, leaving their periodicity unaltered, the cycles do not tighten too fast. The implied parameters are $\phi_1 = 1.6630$ and $\phi_2 = -0.81$. We then set $\sigma_g^2 = 0.2$ as the variance of the error term of the process.

Figure 1: Sine functions



The second dataset aims to capture the widely held view that expansions and recessions are not symmetric phenomena. Expansions tend to be longer and shallower, while recessions can be abrupt, short and deep. To model this we use sine functions. We start from the symmetric function

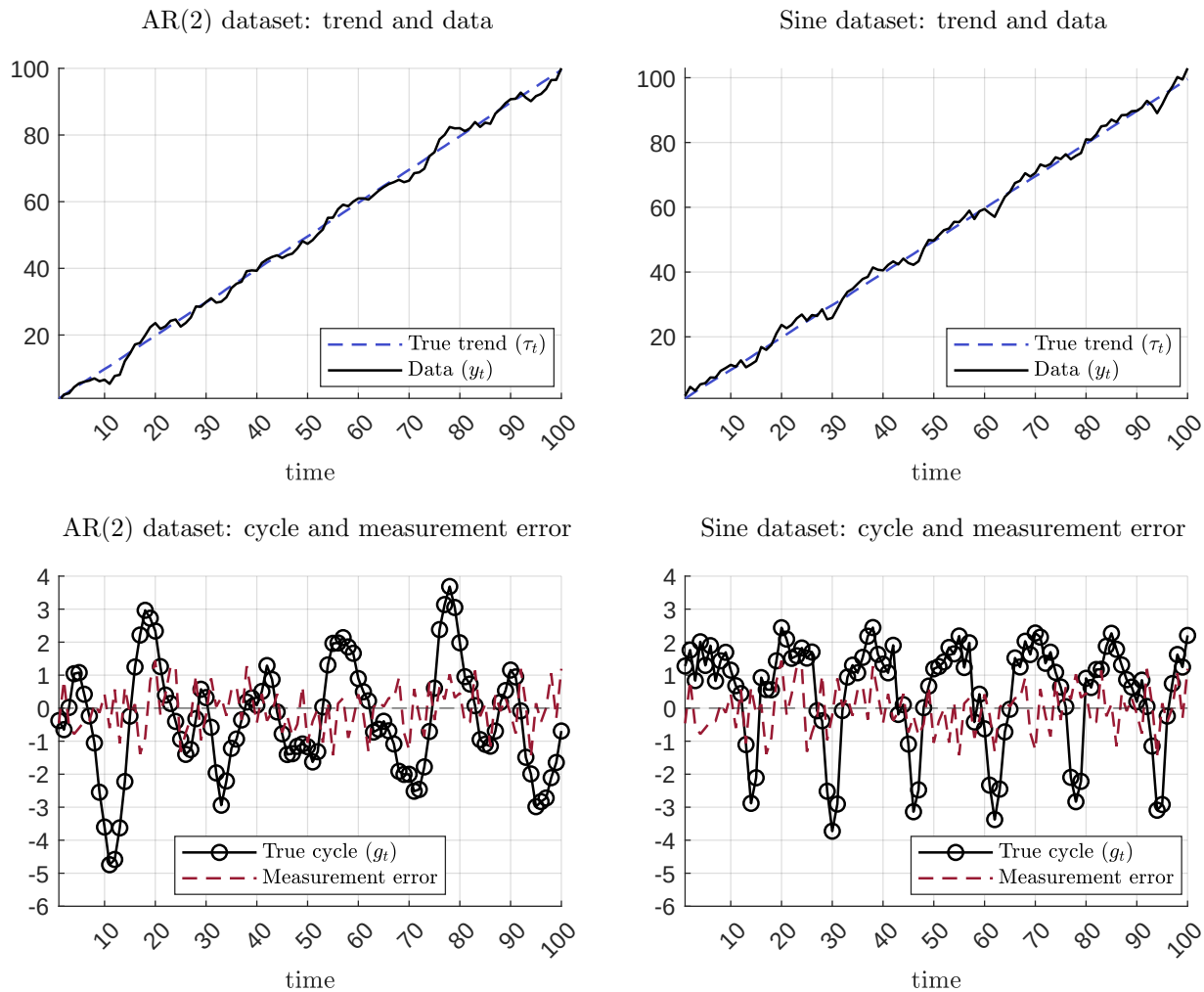
$$\delta_1 \sin\left(\frac{t-\delta_2}{2\delta_3}\pi\right), \tag{11}$$

which equals zero at the generic period $t = \delta_2$, takes δ_3 periods to complete an expansion which peaks at $+\delta_1$, and then takes an additional δ_3 periods to complete a recession down to $-\delta_1$ (left panel of Figure 1). We generalize equation (11) to

$$\bar{g}_t = \begin{cases} \delta_{1E} \sin\left(\frac{t-\delta_2}{\delta_{3E}}\pi\right) & \delta_2 \leq t < \delta_2 + \delta_{3E}, \\ -\delta_{1R} \sin\left(\frac{t-(\delta_2+\delta_{3E})}{\delta_{3R}}\pi\right) & \delta_2 + \delta_{3E} \leq t < \delta_2 + \delta_{3E} + \delta_{3R}. \end{cases} \tag{12}$$

Starting at period $t = \delta_2$, an expansion takes δ_{3E} to complete and peaks at $+\delta_{1E}$, while

Figure 2: Simulated datasets



a recession takes δ_{3R} to complete and reaches at $-\delta_{1R}$ (right panel of Figure 1). By construction, (12) converges to (11) for $\delta_{1E} = \delta_{1R} = \delta_1$ and $\delta_{3E} = \delta_{3R} = 2\delta_3$. In the second dataset we use equation (12) and set the true cycle equal to the realization for $T = 100$ periods from the process

$$g_t = \bar{g}_t + \delta_4 \, \epsilon_t, \qquad \epsilon_t \sim \mathcal{N}(0, 1), \tag{13}$$

with $\boldsymbol{\delta} = (\delta_{1E}, \delta_{1R}, \delta_2, \delta_{3E}, \delta_{3r}, \delta_4)'$ and

$$\delta_{1E} = 2\%, \quad \delta_{1E} = -3\%, \quad \delta_2 = 0, \quad \delta_{3E} = 12, \quad \delta_{3R} = 4, \quad \delta_4 = 0.2. \tag{14}$$

This parametrization implies that, compared to the AR(2) dataset, expansions last 4 periods longer and recessions last 4 periods shorter. The calibration of $(\delta_{1E}, \delta_{1R})$ ensures that also the magnitudes of the cycles are comparable across datasets.

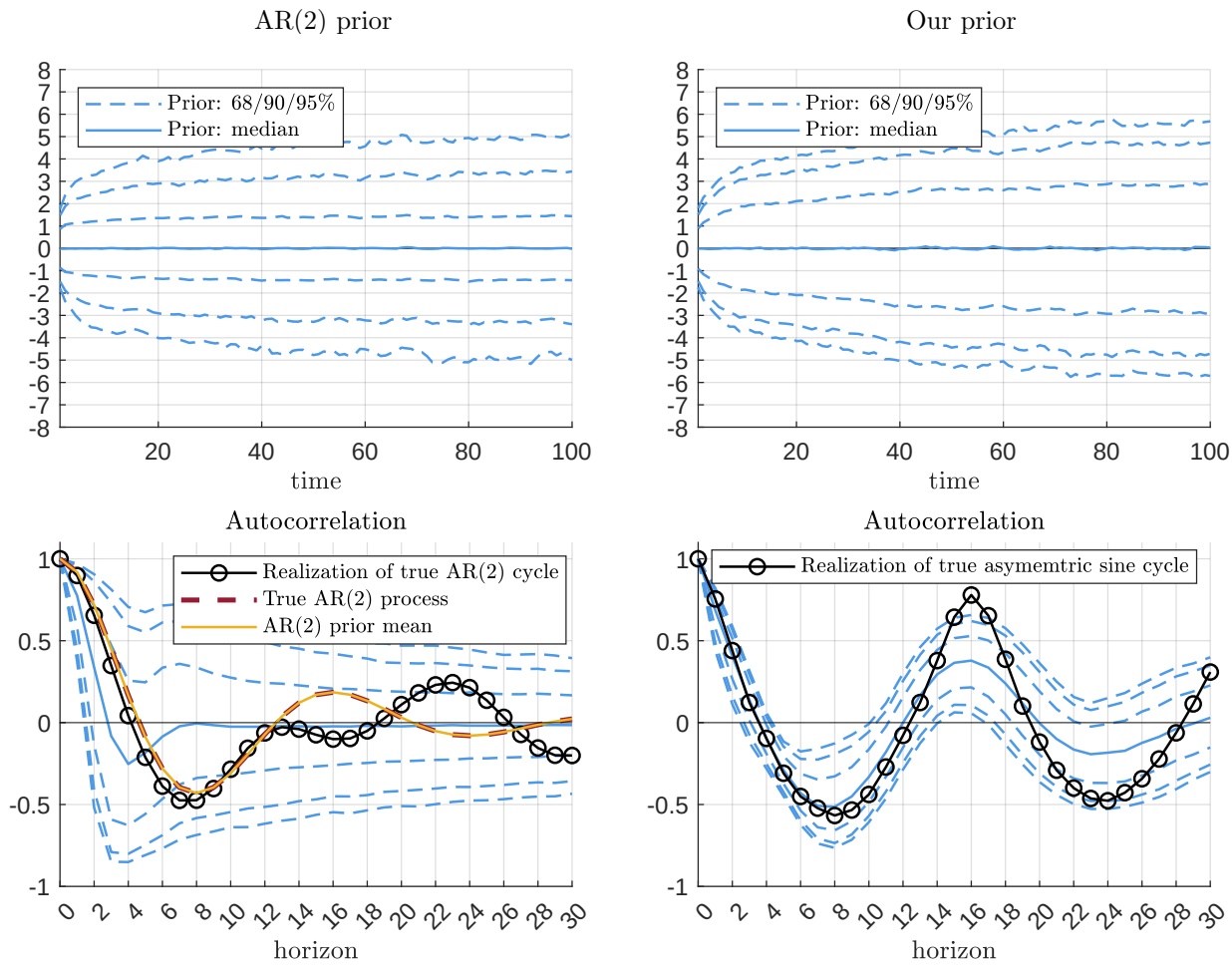
For each dataset, we compute the pseudo (log of) real GDP data as the sum of the true trend and the true cycle. We then add measurement error drawn from a Normal distribution with standard deviation 0.5, ensuring the same draws across datasets to improve comparability. Figure 2 shows the pseudo datasets. The left column refers to the dataset featuring an AR(2) cycle, while the right column refers to the dataset using the asymmetric sine cycle. The top panels show the true trend (blue dashed line) and the dataset (solid black line). The bottom panels show the true cycle (black circled lines) and the measurement errors (red dashed lines). For the exercise in this section it suffices to generate a single draw of each dataset. The Online Appendix reports a more comprehensive exercise with multiple draws of each dataset.

We estimate model (1) using, for both datasets, the same prior beliefs on the variance of the measurement error σ_u^2 and the same prior on the trend $\boldsymbol{\tau}$. For the former, we use an inverse gamma prior centered at 3 and with variance 2. For the latter, we use the random walk prior with drift from equation (2). We use prior $p(\mu, \sigma_\tau^2, \tau_0) = p(\mu)\ p(\sigma_\tau^2)\ p(\tau_0)$, with a Normal prior for τ_0 centered at the first realization of the data and with variance 100, a Normal prior on μ centered at the value consistent with the ratio $\frac{y_T}{y_1}$ and with variance 0.5, and inverse gamma prior on σ_τ^2 centered at 0.03 with variance 0.005.

We then consider two alternative prior beliefs for the cycle. The first prior follows an AR(2) process as in equation (3), and is set to approximate the true AR(2) process of the first dataset. The joint Normal prior on (g_{-1}, g_0) is centered at zero with covariance 0.3 I_2. The joint Normal prior on (ϕ_1, ϕ_2) is centered at the true values, with covariance matrix $0.2 I_2$. The inverse gamma prior on the error term of the AR(2) prior is set to have mean 0.25 and variance 0.065.

The second prior we consider for the cycle is a prior informed by the asymmetric sine data generating process. One could in principle replace the AR(2) model-based prior with equation (12) and make inference on the hyperparameters $\boldsymbol{\delta}$. This approach is computationally more challenging, due to the nonlinearities involving equation (12) with respect to $\boldsymbol{\delta}$ and the lack of conditionally conjugate priors. Our proposed approach does not require sampling $p(\boldsymbol{\delta}|Y)$. Instead, it uses the asymmetric process (12) to *inform* a prior distribution that features a higher degree of tractability. We consider

Figure 3: Cycle: prior distributions



the prior distribution

$$\boldsymbol{g} \sim N(\boldsymbol{\mu}_g, \Sigma_g), \tag{15}$$

which is a simplied specification of the composite prior from equation (9). We set $\boldsymbol{\mu}_g = \mathbf{0}$ to ensure that the first moment of the posterior is not biased towards the truth via the prior mean. As for Σ_g, we set it as follows, using simulations from equation (12):

1. draw δ_{1E} from a continuous uniform distribution with support $[1.8, 2.2]$;
2. draw δ_{1R} from a continuous uniform distribution with support $[2.8, 3.2]$;
3. draw δ_2 from a discrete uniform distribution with support $[-20, 20]$;

4. draw δ_{3E} from a discrete uniform distribution with support $[10, 15]$;

5. draw δ_{3R} from a discrete uniform distribution with support $[1, 5]$;

6. set $\delta_4 = 0.02$

7. draw $\boldsymbol{g}$ from (12) given the above parameters $\boldsymbol{\delta}$;

8. repeat from step 1 a desired number of times;

9. store all draws. Set Σ_g equal to the empirical estimated covariance function as the average covariance across simulations.

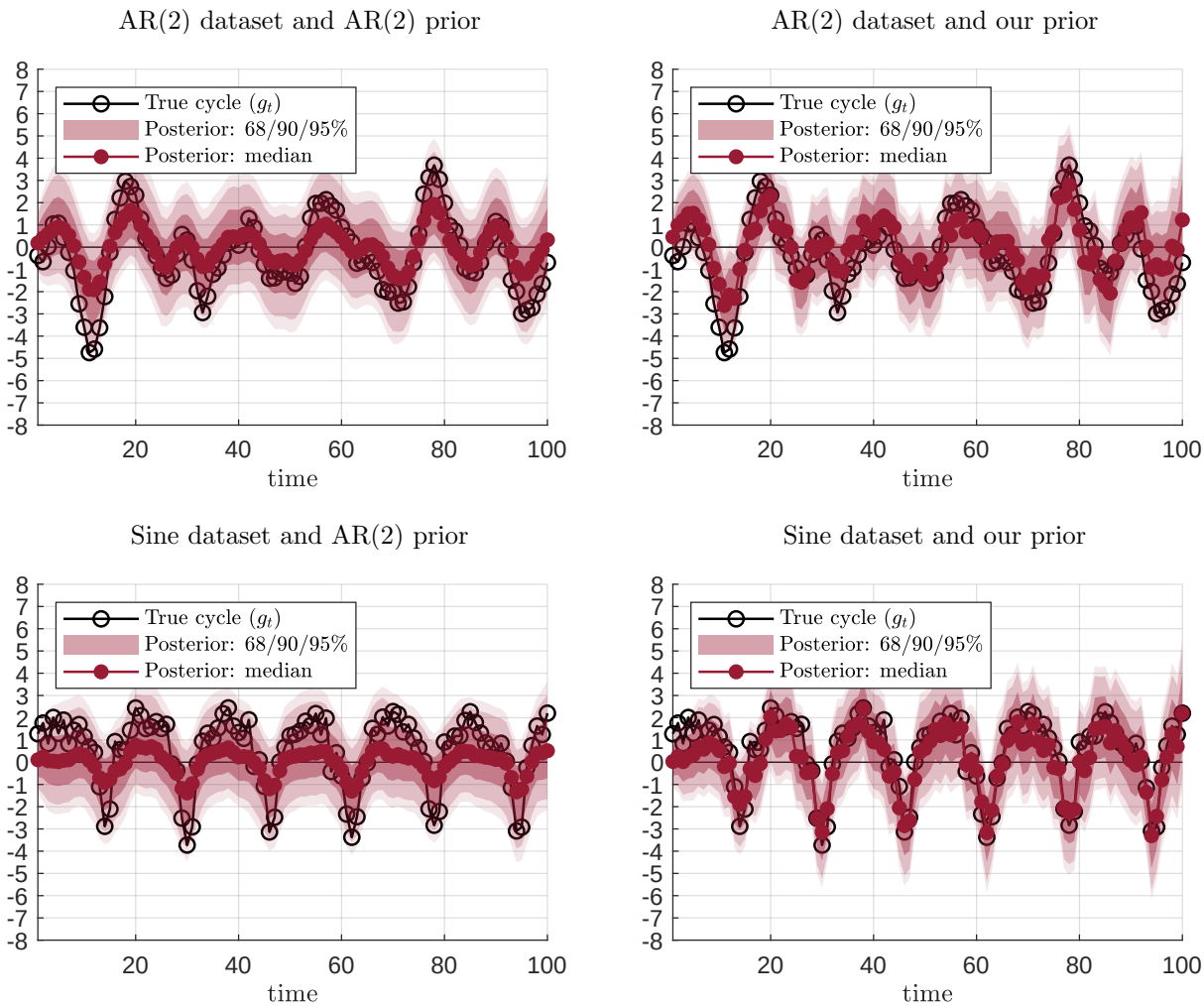
The logic of the calibration of Σ is simple. We aim to incorporate into the prior the true belief that the data generating process generates asymmetric cycles. The randomlization on the parameters $\boldsymbol{\delta}$ is loose but centered around the true parameter values from the data generating process. This replicates the case in which, in applied work, the researcher holds an economic theory that reveals true information about the underlying unobserved output gap.

To further ensure comparability between the AR(2) prior and the asymmetric sine prior, we use the simulations described above to only construct the prior correlation structure rather than the prior covariance. We then compute the prior covariance to ensure that the marginal prior variances over periods $t = 1, .., 100$ are the same as with the AR(2) prior. This implies that the priors are approximately equally tight, but introduce potentially very different information via the correlation structure. This additional step does not affect the results considerably, but simplifies the illustration.

The top panels of Figure 3 report the pointwise prior median and bands associated with the two alternative prior distributions for the cycle. As it is clear from the figure, neither prior introduces a prior mean that reflects any cyclicality, hence neither prior biases the first moment towards the truth. The prior variance of the AR(2) is smaller at shorter horizons, consistent with the fact that the marginal prior for (g_{-1}, g_0) is not the marginal distribution implied by the AR(2) itself. By construction, the width of the two priors is very similar, since only the correlation structure across time different across priors.

The lower panel of Figure 3 helps explore the difference in the two prior distributions by reporting a series of autocorrelation functions up to horizon 20. These are computed either on the realization of the true cycle, the realization of each prior draw, or the population autocorrelation function of the process. In the bottom left panel, the

Figure 4: Cycle: posterior distributions



dashed red line shows the true autocorrelation function of the data generating process using an AR(2) cycle. This coincides with the yellow line, which shows the population autocorrelation of the AR(2) prior distribution for $\boldsymbol{g}$ evaluated at its mean. Since the prior mean for ϕ_1, ϕ_2 was set equal to the true values from the AR(2) dataset, the two lines coincide, and are very close to the autocorrelation function computed on the true realization of the cycle under the AR(2) dataset, marked in the black circled line. All three lines decrease for the first 8 periods and then increase for the subsequent 8 periods, reflecting the fact that it takes 8 periods for the AR(2) process to complete a full expansion or a full recession. Yet, the AR(2) prior fails to introduce this feature of the data despite a fairly tight prior on (ϕ_1, ϕ_2). This is clear from the blue lines, which report pointwise mean and percentiles that decease at horizon 4 rather than

8. The bottom right panel of Figure 3 shows the autocorrelations for our prior. The black circled line shows that the autocorrelation function of the true asymmetric sine cycle is very different from the AR(2) process. But contrary to the AR(2) prior, our prior introduces patterns that are consistent with the data generating process.

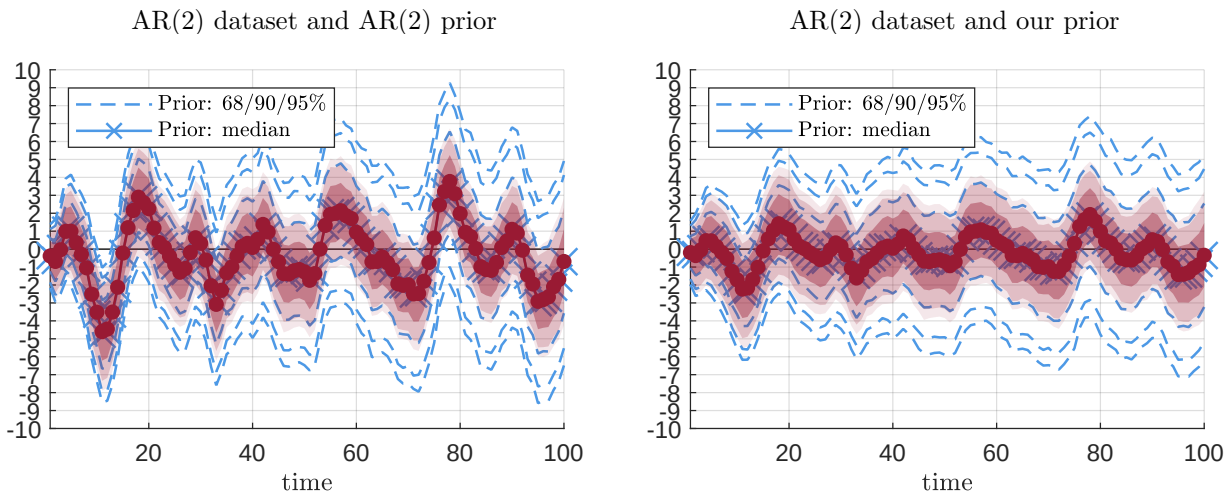
Figure 4 shows posterior distributions on the cycle (see the Online Appendix for the results for the trends). The figure reports pointwise median and bands associated with all four combinations of the two datasets and the two priors. The left column of Figure 4 reports the results using the AR(2) the prior. As is clear from the figure, inference is very successful when the underlying true cycle is indeed an AR(2) process. However, the results are less satisfactory when the true cycle is the asymmetric sine function, despite on average similar patterns between datasets. In fact, the bottom left figure shows that the sampler correctly infers only the qualitative feature of the cycles, uncovering the true pattern of expansions and recessions. Yet, it fails to capture the true cycle quantitatively, underestimating the cyclical dynamics by around 50%. The results are different for our prior, as shown on the right hand side of the figure. Our prior successfully recovers the true process under both datasets.

The results from Figure 4 suggest that prior distributions can have a significant role in the estimation of latent variables. Indeed, by construction, the true trend and the true measurement error was the same across datasets, hence, within dataset, the only factor driving the difference in the posterior is the difference in the prior distributions on $\boldsymbol{g}$. We further support this illustration with an additional exercise. Consider for simplicity only the AR(2) dataset. Replace the random walk prior with a joint Normal prior

$$\boldsymbol{\tau} \sim \mathcal{N}(\boldsymbol{\mu}_\tau, \Sigma_\tau), \tag{16}$$

where we set Σ_τ to feature the same prior correlation structure as the random walk prior used so far, with prior marginal variances for each period set equal to the average variance of the random walk prior (this parametrization is inconsequential for the illustration). For both cases below, set $\boldsymbol{\mu}_\tau = \boldsymbol{y} - \boldsymbol{\mu}_g$. For Case a), set $\boldsymbol{\mu}_g$ equal to the true cycle, while for Case b) set it equal to half of the cycle. The prior and posterior distributions are reported on the left and right panel of Figure 5. It is clear from the figure that in neither case the first prior moment is updated, as prior and posterior pointwise median coincide under both cases. The only effect of introducing the data to the analysis is to tighten uncertainty. The result is due to the lack of identification of trend and cycle. Since by construction the prior distributions on $\boldsymbol{\tau}$ and $\boldsymbol{g}$ sum up to the data, the data cannot update the prior means.

Figure 5: Cycle: illustration of lack of identification



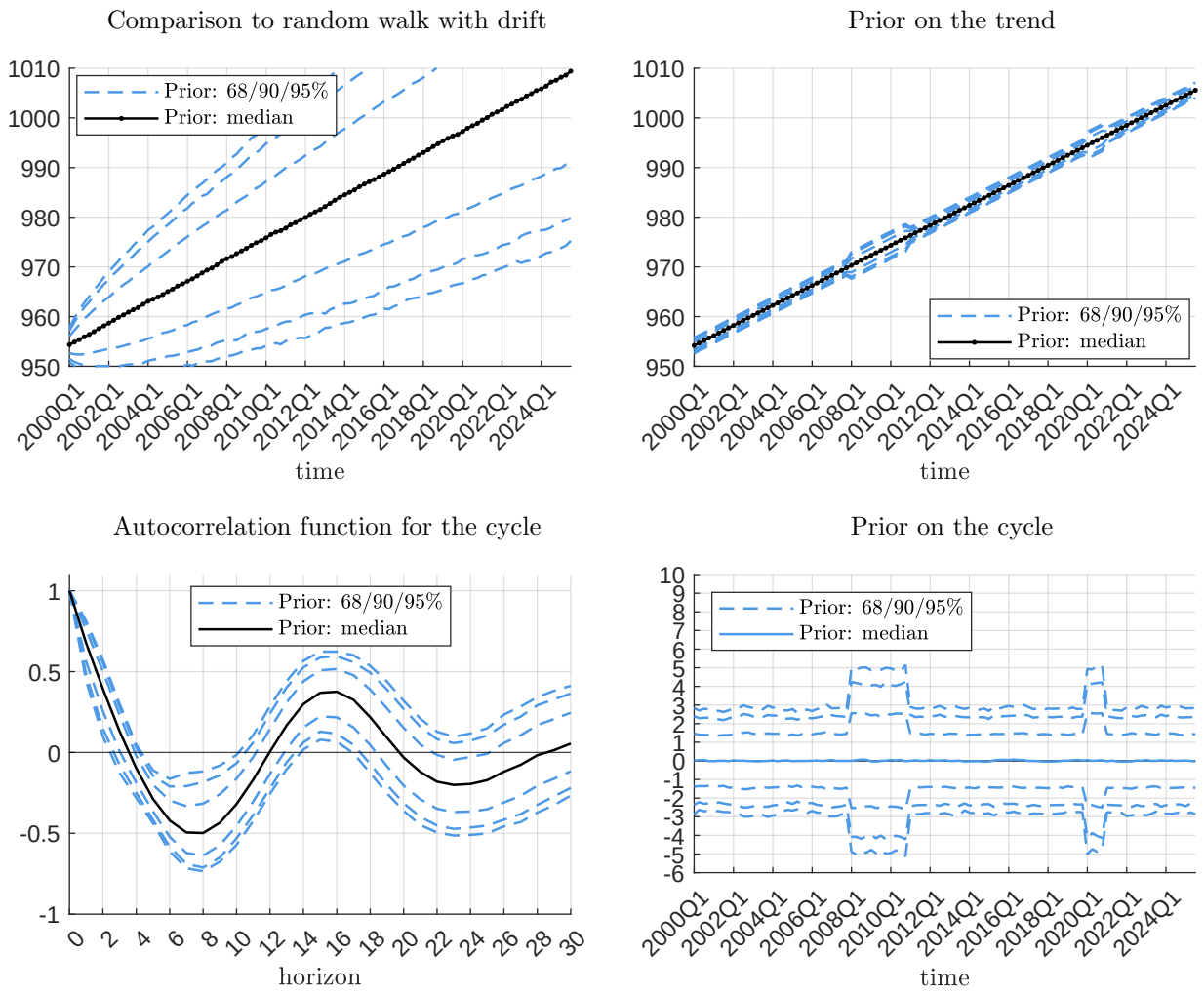
The take away from both exercises is not that our prior is superior to the AR(2) prior. The take away is simple: prior distributions matter. This result is not due to the length of the dataset, but to the nature of the decomposition exercise. Because of this identification problem, it is important to carefully select prior distributions on $(\boldsymbol{\tau}, \boldsymbol{g})$, potentially combining statistical information with theory-based beliefs.

4 Application to US data

We apply our methodology to estimate the evolution of potential real GDP and the output gap in the US in the period between 2000Q1 and 2025Q3. We use our methodology to explore the following question: is there any sign that some of the drop in economic activity following the Great Financial Crisis and/or following the Covid pandemic was at least in part associated with a drop in potential output? Alternatively, was it only due to a drop in the temporary level of output, namely a widening of the output gap?

We use our flexible prior approach to specify a prior distribution on both the trend and the cycle. We use a joint Normal prior for the trend, centered at $\boldsymbol{\mu}_\tau$ and with covariance matrix Σ_τ. Following an Empirical Bayes approach, we set $\boldsymbol{\mu}_\tau$ equal to the linear fitted line computed on the data. This ensures a prior distribution for $\boldsymbol{\tau}$ centered at a reasonable value. For the variance, our starting point is the covariance structure of a random walk prior with drift (equation 2) which conditions on $\tau_0 = y_1$, sets μ

Figure 6: Prior distributions on trend and cycle



consistent with the slope y_T relative to y_1, and sets $\sigma_\tau^2 = 3$. The prior distribution implied by this prior is shown in the top left plot of Figure 6, and shows the typical explosive variance as the horizon increases.

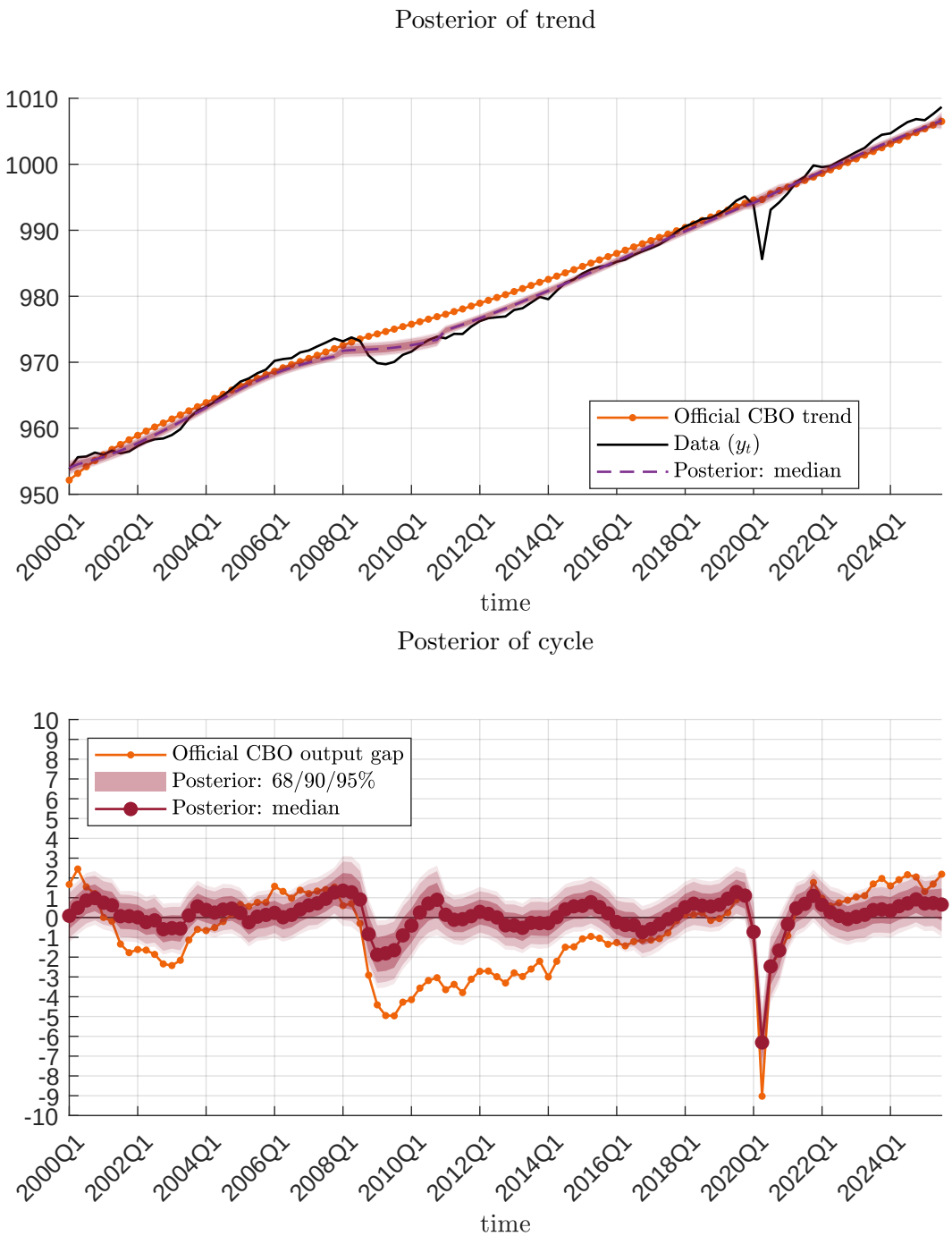
Starting from the prior variance of the random walk with drift, we store the implied correlation structure and introduce it into our prior covariance Σ_τ. We then set the marginal variances for each period to ensure a reasonable variance, avoiding the explosive prior variance of the random walk. The prior mean of the trend implies an annualized quarter-on-quarter growth rate of the trend of around 2.1%. We then set the variances in Σ_τ to ensure that 95% prior mass in each period lies between $\pm 5\%$ from the unconditional mean. This calibration ensures that the trend can still fluctuate, but in a relatively smooth way. Last, we scale up by a factor of 3 the variances corresponding to years 2008, 2009, 2010, 2020: the first three in correspondence to the Great Financial Crises, the latter in correspondence to the Covid Pandemic. The implied prior is reported in the top right plot of Figure 6.

As for the prior distribution on the cycle, we use a joint Normal prior distribution with mean $\boldsymbol{\mu}_g$ and covariance Σ_g. We set $\boldsymbol{\mu}_g = \mathbf{0}$, to avoid biasing the results via its first moment. We then follow the simulation exercise and set Σ_g using simulations from equation (12), modelling the belief that expansions last longer than recessions, while recessions are potentially fast, deep and abrupt. We use the same simulations of the parameters $\boldsymbol{\delta}$ from equation (12) as in the simulation. Last, consistent with the prior on the trend, we scale up by a factor of 3 the marginal variances corresponding to years 2008, 2009, 2010, 2020. The distribution for the cycle and the corresponding autocorrelation function are shown in the bottom panels of Figure 6.

We set the prior distribution for the variance of the measurement error, σ_u^2 using an inverse gamma distribution. We fix its hyperparameters to ensure a mean and variance of the unconstrained distribution equal to 0.25 and 5, respectively. We then truncate the distribution to have positive mass between 0 and 1. This prior ensures that most of the evolution of the data is picked by the posterior distribution of the trend and/or the cycle, rather than by the iid measurement error.

Figure 7 reports the posterior distribution for both the trend (top panel) and the cycle (lower panel). Pointwise median and bands are reported, together with the evolution of the data (100 times log of real GDP, black solid line in the top panel) and the official CBO estimate of both the trend and the cycle (orange dotted line). Figure 7 clearly shows that the Great Financial Crisis is not exclusively interpreted by the model as an opening of the output gap. Instead, the model partly interprets

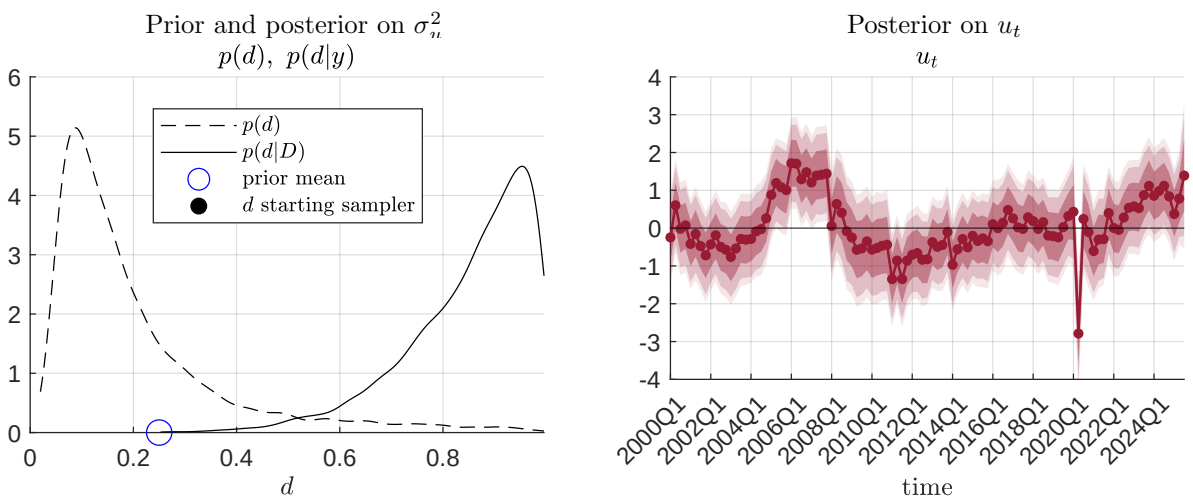
Figure 7: Prior distributions on trend and cycle



it as a slow but persistent decrease in the potential level of output, whose decrease starts before the first drop in real GDP. Because the official CBO estimate reports an almost non existent decline in the trend, our estimated output gap is only qualitatively similar to the CBO output gap, but quantitatively smaller.

The results for the Covid pandemic period is, instead, very different. As explained above, the variance of both the trend and the cycle, was scaled up by the same factor in the four quarters of 2020. Yet, the model can successfully interpret the drop of economic activity as exclusively as a drop in the output gap, while leaving the estimated potential level of output unaffected.

Figure 8: Prior distributions on trend and cycle



As clear from Figure 7, our estimate of the output gap in the Covid period is smaller than the CBO estimate, while the estimated trends are very similar. Since the estimated CBO trend and cycle sum up to the data, the difference in the results for the output gap are interpreted by the model as a measurement error. Figure 8 reports this by showing, on the left, the marginal prior and posterior distributions on σ_u^2, while on the right the pointwise bands for the posterior draws of u_t, which equal, by construction, $y_t - \tau_t - g_t$. As clear from the figure, the model partly interprets the strong economic activity preceding the Great Financial Crisis as a positive measurement error, as well as 3% of the CBO output gap in 2020Q2.

5 Conclusions

Decomposing GDP into potential output and the output gap is fundamentally an exercise in identification. Because both components are latent, any empirical decomposition necessarily embeds restrictions on the joint dynamics of trend and cycle. The choice and strength of these restrictions have direct implications for inference and policy analysis.

This paper develops a general econometric framework that makes these restrictions explicit and continuously tunable. By specifying priors directly over the joint distribution of the latent trajectory, rather than through a particular state evolution equation, the framework nests standard unobserved-components models while enlarging the set of admissible dynamics. Identification is therefore decoupled from a specific parametric law of motion and instead implemented through mean and covariance restrictions on the full path of the latent variables.

The simulation evidence shows that our approach is sufficiently flexible to potentially introduce non-linear and asymmetric features into the prior moments, without having to estimate non-linear models. Rather, only simulation from such model is required to inform our prior moments. The empirical application shows the Great Financial Crisis was associated with a significant drop in potential output. By contrast, the Covid pandemic is not found to have had an effect on potential output, and is interpreted by the model as a combination of a widening output gap and a measurement error.

Although the analysis of our paper focuses on potential output and the output gap, the framework applies more broadly to the estimation of latent "star" variables. Our framework can be extended to time-vaying paramer models in which economic theory informs the potential evolution of the parameters.

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